

Printed surface texture and the electrical output of a small vertical-axis wind turbine rotor

Wind-tunnel results, August–October 2026

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Data URI_VAWT_Roughness_Data_2026-10-01.zip (accompanies this report)

Summary

Bottom line. Tested on one day against the un-textured rotor, the rotors printed with 0.101 mm and 0.202 mm of fuzzy-skin texture produced +12.8% and +16.2% more peak electrical power over 10.2–38.0 m/s. Allowing for mounting variation, estimated from the earlier runs, the 95% intervals are [+5, +21]% and [+9, +24]%; both gains are resolved. The 0.050 mm rotor’s +1.8% is not. The gain sits in the open-circuit voltage, consistent with the rotor turning faster in the same wind; rotor speed was not measured. In the point estimates power rises with fuzzy-skin thickness, not with measured roughness: Ra is 8.2–10.3 μm on the three textured rotors and 12.7 μm on the un-textured one.

Table 1. Result of the 1 October session, against the un-textured rotor.

Rotor	Fuzzy-skin thickness	Pa (μm)	Ra (μm)	Run 1 / 2 (%)	Change, 95% CI ^a (%)	Run-to-run ^b (%)	P_{max} , 38.0 m/s (W)
No texture	none	19.0	12.7	−0.4 / +0.4	reference	—	3.67
Ra 20	0.050 mm	13.9	9.4	+2.3 / +1.3	+1.8 [−5, +9]	[+0.1, +3.4]	3.78
Ra 40	0.101 mm	13.2	8.2	+12.4 / +13.1	+12.8 [+5, +21]	[+11.0, +14.6]	4.30
Ra 80	0.202 mm	28.7	10.3	+15.7 / +16.8	+16.2 [+9, +24]	[+14.4, +18.1]	4.53

Change: geometric mean over the 14 fan set points of the ratio to the un-textured rotor, averaged over the rotor’s two runs. Each rotor was mounted once and swept twice (back to back, except the no-texture runs, 60 minutes apart). ^aAllows for mounting variation, estimated from the August–September runs ($\sigma_m = 1.5\%$, 3 degrees of freedom; Section 2.3). ^bRun-to-run repeatability only ($\sigma = 0.6\%$, 4 degrees of freedom). Pa, Ra: Keyence VR-6000 profilometer (Section 1.1).

- **Steps.** Of the three steps in thickness, only 0.050 to 0.101 mm is resolved (its 95% interval excludes zero): +10.8% [+4, +19] (Section 3). Ra 40 and Ra 80 are above the un-textured rotor at all 14 wind speeds.
- **Mechanism.** Open-circuit voltage, a proxy for free-running rotor speed, rises by +6.3% and +8.4% for Ra 40 and Ra 80, while source resistance shows no detectable change. Squared, the voltage change implies +13.0% and +17.6% in power.
- **Reproducibility.** Each textured rotor’s 1 October level is +1.1%, +3.3% and +1.5% above its single run of 20 August–1 September, made in a different order.
- **Caveats.** Each rotor was mounted once on 1 October, so mounting variation comes from the earlier runs. The rotors were run in order of increasing texture (Section 3.4 gives the evidence

against drift). The un-textured rotor was printed with 0.2 mm layers and the Ra 20 and Ra 40 sets with about 0.1 mm, so the reference differs in more than texture. There is one print per texture level. The rig measures electrical power, not the aerodynamic power coefficient.

Responses to open requests.

- *Comparison with the initial test (Prof. Jeong)*. Section 4: the printed rotors perform roughly like the initial reference; the rotor and processing differ.
- *Which blade exactly (Prof. Jahangiri)*. Table 2 lists what is recorded and what was measured for each set.
- *Fault-testing plan (Prof. Jeong)*. Section 6.1 proposes a starting point to agree before any fault blades are printed.

A fifth blade set was also run on 1 October. Its print settings could not be confirmed, so it is not reported.

Acknowledgement. Taegu Kang co-ran the July and August tests, brought the electronic load for the 1 October session, and shared the complete raw export of the 27 July test.

1 Test articles

All rotors share one geometry (v1): three straight, untwisted blades at radius $R = 0.1016$ m with span $H = 0.2451$ m, so the swept area is $2RH = 0.0498$ m². The blade section is a thin cambered plate, not an airfoil: a 1.79 mm wall on a 48 mm chord with 183° of turning and square edges, closer to a half-cylinder bucket than to a wing. The chord Reynolds number spans 3.3×10^4 to 1.2×10^5 .

The sets differ in the slicer’s fuzzy-skin thickness, which randomly displaces the outer wall and doubles from one textured set to the next (Table 2). The labels Ra 20, Ra 40 and Ra 80 are names, not measurements. They come from roughness targets suggested when the prints were planned (4 August), extended by doubling. Fuzzy-skin thickness is the controlled variable and is used throughout.

Table 2. The blade sets. Layer period: dominant spanwise wavelength of the measured surface. Run notes record intent and are not reliable for print settings.

Rotor	Fuzzy skin	Layer period	Record
No texture	none	203 μ m	No fuzzy skin; the period implies 0.2 mm layers. Probably the set tested in the Jeong lab on 27 July (to be confirmed). One blade broke at the end of the second run; $P_{\max}$ at 1800 rpm matches the first run to within 0.1%.
Ra 20	0.050 mm	102 μ m	Not in the slicer file. Printer’s account: garage Bambu, 0.4 mm nozzle, 0.08 mm layers. Run notes: “PETG, 0.2mm”. The period indicates about 0.1 mm layers.
Ra 40	0.101 mm	102 μ m	Slicer file: plate “40 RA”. Printer’s account: same batch as Ra 20. Run notes: “PETG, 0.2mm layer”. Carried a small magnet on one blade for a reed-switch sensor on 1 September (not recorded for 1 October).
Ra 80	0.202 mm	—	Slicer file: plate “80 RA”. Run notes: “PETG, 0.2mm layer, 0.2mm nozzle”. No distinct layer period; the fuzz dominates.

Slicer file. The slicer project `turbine_default.3mf` (Bambu Studio, saved 17 August) holds three plates, each with one blade ($24.5 \times 48.0 \times 245.1$ mm, the v1 blade; a rotor takes three). Two are reported sets: 0.101 and 0.202 mm (plates “40 RA” and “80 RA”); the third holds a set not reported here. The profile is a Bambu Lab P1S with a 0.2 mm nozzle, 0.10 mm layers and PLA. Fuzzy skin is painted on both faces of each blade (96.5% of the surface area; the square edges and end caps are plain) at a 0.2 mm point distance. The 0.10 mm layers match the measured period of Ra 40. The file holds no 0.050 mm or un-textured blade. Its material (PLA) conflicts with the run notes (PETG), and its nozzle with the printer’s account (0.4 mm); it does not record which prints were made from it.

1.1 Surface measurement

On 1 October each set was scanned with a Keyence VR-6000: one 1.95×1.43 mm field per set (two for Ra 80) at $1.85 \mu\text{m}$ per pixel (Fig. 1). Profiles run spanwise, across the layer lines, and are levelled. Pa is the arithmetic mean deviation of the levelled profile; Ra is the same after an ISO 16610-21 Gaussian filter with $\lambda_c = 0.25$ mm (Appendix B).

- **The un-textured surface is not smooth.** Its layer lines form a regular $203 \mu\text{m}$ ridge pattern, which gives it the highest Ra of the four ($12.7 \mu\text{m}$).
- **Ra 20 and Ra 40 are nearly indistinguishable:** Pa 13.9 and $13.2 \mu\text{m}$, Ra 9.4 and $8.2 \mu\text{m}$, on the same layer period.
- **Ra 80 has about twice the Pa** ($28.7 \mu\text{m}$). Its two fields differ by $4.8 \mu\text{m}$ (17%), so one field is a coarse sample.

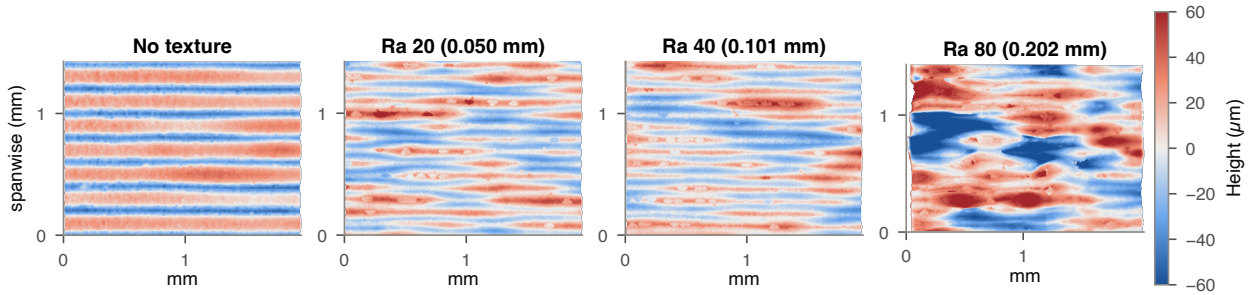


Figure 1. Keyence height maps, one field per set, each column levelled. Layer lines run horizontally; profiles for Pa and Ra run vertically (spanwise). Colour clipped at $\pm 60 \mu\text{m}$.

2 Method

2.1 Rig and protocol

A Python host commands the tunnel fan through a Portenta Machine Control and an ABB ACS550 drive (Fig. 2). The generator feeds a three-phase bridge rectifier and a Chroma 63004 DC electronic load, which sets the current and measures terminal voltage and current. Wind speed is the tunnel calibration $v = 0.02132 N_{\text{fan}} - 0.424$ (m/s, rpm; $R^2 = 0.9996$) applied to the commanded fan speed.

Each sweep visits 14 fan set points from 500 to 1800 rpm (10.2–38.0 m/s). At each, the load raises its constant-current demand in 1 s dwells (20 mA at 1800 rpm, scaled with v^2 below) and stops when power falls below 80% of its running maximum. A sweep takes about 8 minutes. Every run carries protocol fingerprint 94bed28333f7.

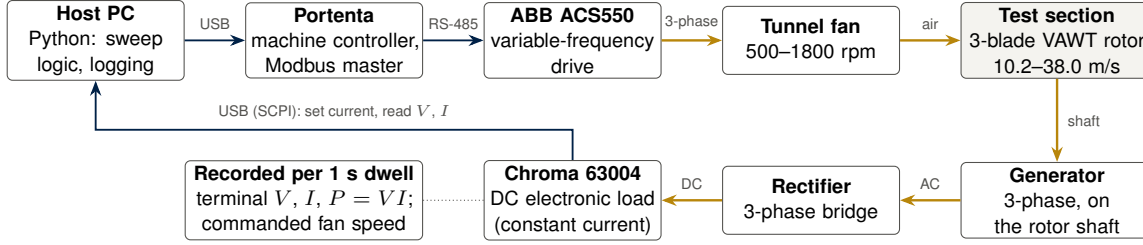


Figure 2. Measurement and control chain. Rotor speed was not recorded.

2.2 The 1 October session

Each rotor was mounted once and swept twice without being removed, in the order no texture, Ra 20, Ra 40, Ra 80, from 15:50 to 18:24 (first to last load step). The two no-texture runs started 60 minutes apart; the other pairs ran back to back. 110 of 112 set points ended on the roll-off rule. The other two ended on the load’s voltage-floor cut-out after power had already fallen well below its peak: the no-texture rotor run 1 at 500 rpm (49%) and Ra 20 run 2 at 600 rpm (57%) of peak. In every ladder power fell after the peak, so every $P_{\max}$ lies inside the ladder. Air temperature and pressure were not recorded.

2.3 Analysis

- **Peak power** is the largest 1 s dwell. A parabolic fit through the peak gives nearly identical changes (Appendix B).
- **Comparison.** For each run, y is the mean over the 14 set points of $\ln(P_{\max}/P_{\text{ref}})$, where P_{ref} is the geometric mean of the two no-texture runs at that set point. The run-to-run standard deviation, pooled within rotors, is 0.6% (4 degrees of freedom). No rotor was remounted, so this is repeatability only.
- **Mounting variation** is estimated from the August–September runs, each a separate mounting: their differences from the 1 October means imply a mounting standard deviation $\sigma_m = 1.5\%$ (3 degrees of freedom), which also absorbs any day-to-day change. It is assumed the same for the no-texture rotor, which has no earlier run. The headline 95% intervals include it (Appendix B). Print-to-print variation is not included.
- **Thévenin decomposition.** At each set point the ladder follows $V = V_{\text{oc}} - IR_{\text{int}}$ ($r^2 \geq 0.965$ in every run). V_{oc} rises with free-running rotor speed, offset by the rectifier drop; R_{int} combines the electrical path with the rotor’s slowing under load. At the peak, $P_{\max} \approx V_{\text{oc}}^2/4R_{\text{int}}$.

3 Results

3.1 Peak power

Every run of every textured rotor lies above both no-texture runs, and the two runs of any rotor differ by less than 1% (Fig. 3, Table 4). The August–September runs, each on another mounting and day, differ from the 1 October means by 1–3%. Allowing for that mounting variation, the changes between neighbouring thicknesses are:

- no texture to 0.050 mm: +1.8% [−5, +9];
- 0.050 to 0.101 mm: +10.8% [+4, +19];
- 0.101 to 0.202 mm: +3.0% [−4, +10].

Only the middle step is resolved, meaning its 95% interval excludes zero. Ra 80 was above Ra 40 in both campaigns (+4.9% in August–September, +3.0% on 1 October), but the 1 October interval includes zero. The Ra 20-to-Ra 40 step is the cleanest texture comparison in the data: by the printer’s account the two sets share a print batch, they share a measured layer period, and they differ otherwise only by the magnet on one Ra 40 blade.

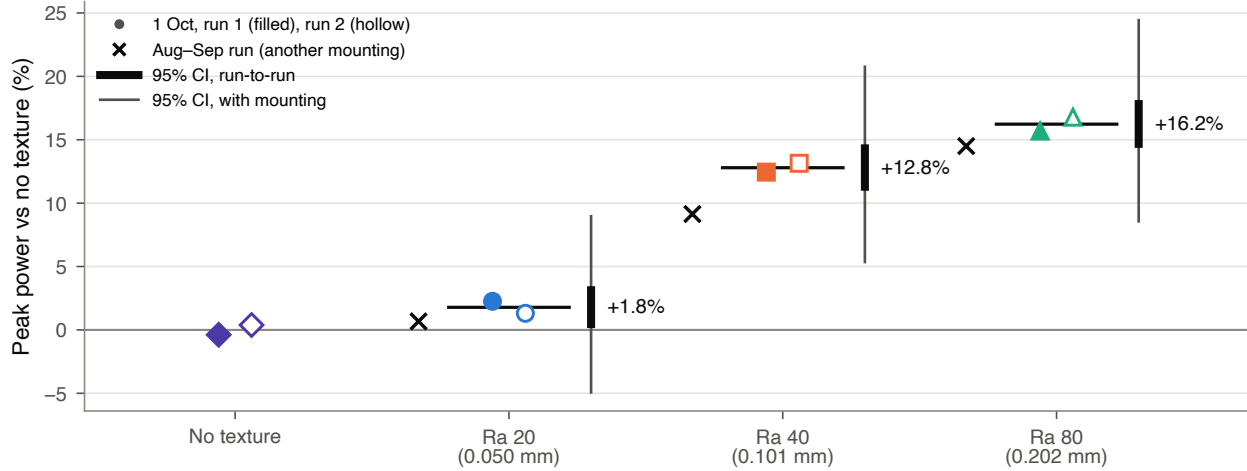


Figure 3. Peak power of every run relative to the un-textured rotor on 1 October (mean over 14 set points). Filled and hollow: the two 1 October runs on one mounting. Crosses: the August–September run, another mounting and day. Bars: rotor mean with 95% intervals, run-to-run (thick) and with mounting variation (thin).

3.2 Variation with wind speed

Ra 40 and Ra 80 are above the un-textured rotor at all 14 set points, and Ra 20 at 9 (Fig. 4). The gain is not uniform. All three textured rotors gain most at 23–25 m/s (Ra 80 +36.2%, Ra 40 +22.4%) and least near 12–14 m/s. Because all three rise together, part of that peak may be a dip in the reference curve rather than a texture effect; these data do not separate the two.

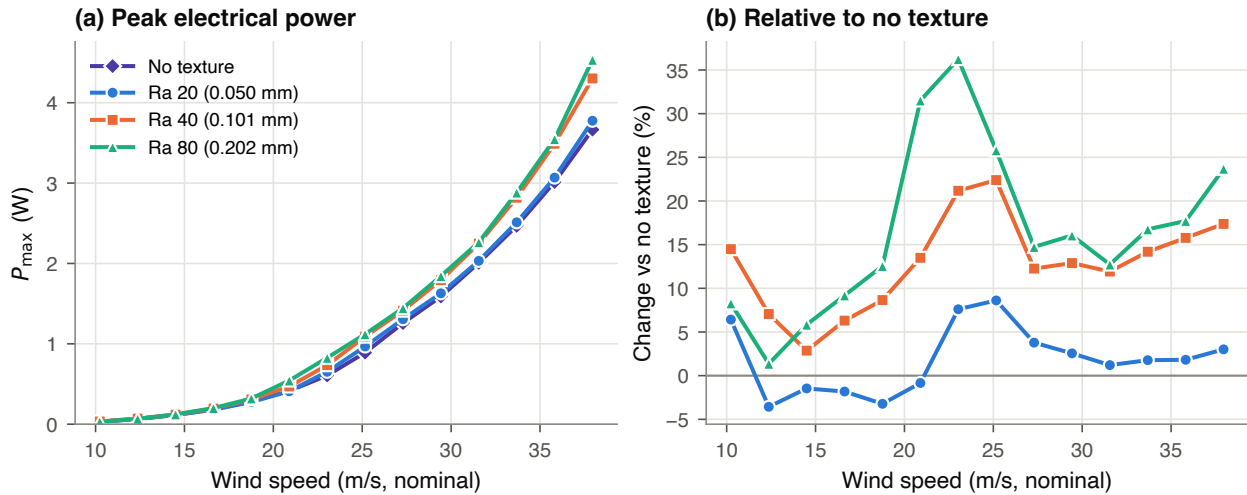


Figure 4. (a) Peak power against nominal wind speed, geometric mean of each rotor’s two runs. (b) The same relative to the un-textured rotor.

3.3 Where the extra power comes from

Relative to the un-textured rotor (Fig. 5; mean over set points, $\pm 95\%$ half-width across set points):

- **Open-circuit voltage** rises by $+1.1 \pm 1.2\%$, $+6.3 \pm 1.4\%$ and $+8.4 \pm 2.5\%$ for Ra 20, Ra 40 and Ra 80. It is higher at 9, 14 and 14 of 14 set points.
- **Source resistance** changes by $-0.2 \pm 1.5\%$, $-0.3 \pm 2.0\%$ and $-0.0 \pm 2.8\%$: no detectable change.

The gain therefore sits in V_{oc} : with R_{int} unchanged, its square gives $+2.3\%$, $+13.0\%$ and $+17.6\%$ against $+1.8\%$, $+12.8\%$ and $+16.2\%$ measured. This decomposes the same ladders; it is not an independent measurement. A higher V_{oc} in the same wind means faster free-running rotation, from more aerodynamic torque, less parasitic torque, or both. Rotor speed was not measured.

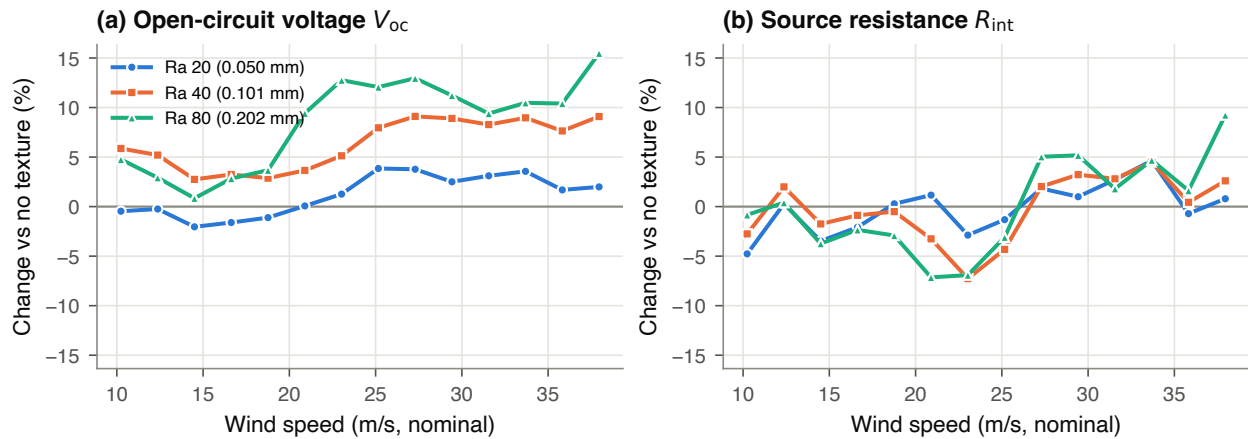


Figure 5. Thévenin decomposition relative to the un-textured rotor. (a) Open-circuit voltage. (b) Apparent source resistance.

3.4 Run order and drift

The rotors were run in order of increasing texture, so a steady drift through the afternoon would mimic a texture effect. Three observations argue against a large drift; none excludes one of a few percent.

1. The two no-texture runs, 60 minutes apart, differ by $+0.8\%$, within run-to-run scatter. Extrapolated over the 154-minute session that is about 2.0% , but one pair cannot exclude more.
2. The fan drew the same current at every set point in all eight runs, within one 0.1 A count of the drive's reading (13.6 A at 1800 rpm). Fan load depends on air density, so a large density change is unlikely. This is not a calibrated density measurement.
3. The textured rotors are $+1.1\%$, $+3.3\%$ and $+1.5\%$ above their August–September single runs. Those runs were made over 12 days in the order Ra 20, Ra 80, Ra 40, and put Ra 40 and Ra 80 $+8.4\%$ and $+13.7\%$ above Ra 20, against $+10.8\%$ and $+14.2\%$ on 1 October.

4 Relation to the Jeong-lab tests

Two earlier tests were recorded on the Jeong-lab DAQ: the initial reference (5 June, probably the original rotor) and a no-texture print (27 July). Fig. 6 compares both with the rig's un-textured rotor at matched fan set point.

- **5 June.** At 800–1700 rpm the values differ from the rig’s un-textured rotor by -4.7% to $+35.0\%$ (geometric mean $+8.8\%$): inside the band of the textured rotors at 5 of 10 set points, above it at 2 and below it at 3. The rotor differs, and the June values are maxima of short averages, which read high against the rig’s 1 s dwells. This shows only that the printed rotors perform roughly like the initial reference.
- **27 July.** The lab’s table reports maxima of raw 360 Hz samples, which collect current-sensor noise (about $0.11 \text{ A} \times V$). At 500–700 rpm there was no load current (Appendix C). Reprocessed on the rig’s basis, the July values differ from the rig’s un-textured rotor by -33.1% to -5.3% , below it at all 9 usable set points. The gap cannot be read as a rotor difference. The load was held below the power peak, so the July values are lower bounds, and the current scale cannot be checked from the file. The free-running voltages, which need no current calibration, differ by at most 15%: 3.11, 3.84 and 5.39 V in the lab against 2.89, 4.52 and 5.67 V on the rig at 500, 600 and 700 rpm.

Recording the lab DAQ in parallel with the Chroma load during one rig sweep would calibrate the current channel and give the rig the rotor speed it lacks.

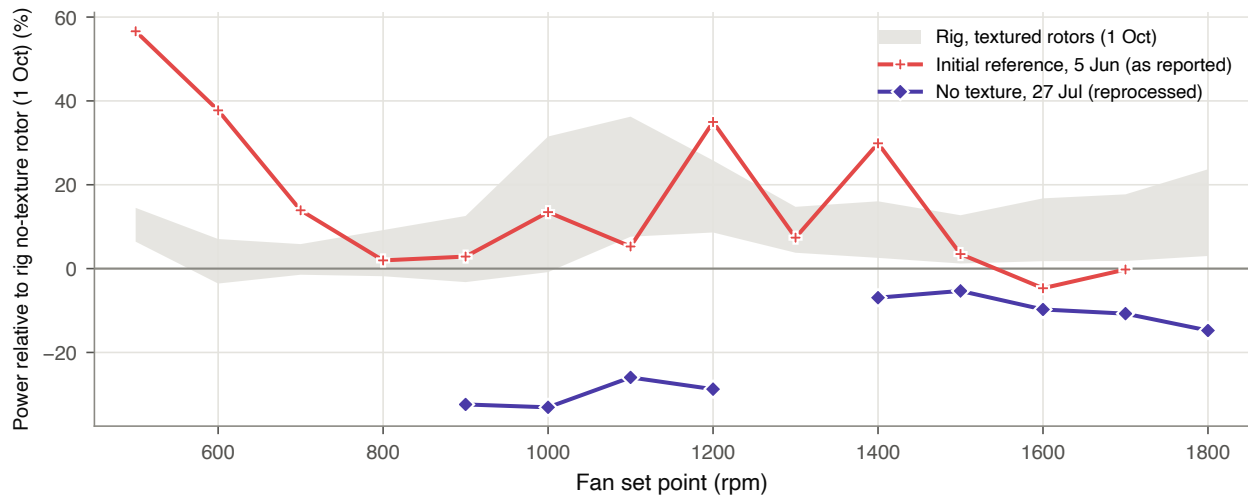


Figure 6. Jeong-lab tests relative to the rig’s un-textured rotor (1 October) at the same fan set point. Grey band: the rig’s textured rotors. 27 July: reprocessed (Appendix C); 500–800 rpm omitted (load off, or far below the power peak) and 1300 rpm (window spans a speed step). 5 June: as tabulated by the lab.

5 Discussion and limitations

What is established. The rotors with 0.101 and 0.202 mm of fuzzy skin produce more peak electrical power than the un-textured rotor, by margins that mounting variation and day-to-day change do not explain. The gain sits in the open-circuit voltage, consistent with faster free-running rotation. Whether 0.050 mm helps, and whether 0.202 mm beats 0.101 mm, is not resolved.

Which surface property matters. R_a at $\lambda_c = 0.25 \text{ mm}$ does not rank the rotors by power: the un-textured rotor has the highest R_a and the lowest power, and $R_a 40$ the lowest R_a and the second-highest power. P_a separates only $R_a 80$. The $R_a 80$ map shows features up to about 1 mm across (Fig. 1), comparable to the 1.43 mm field, so a 0.25 mm cut-off removes much of the texture. Longer profiles are needed before any roughness parameter can be tied to performance.

Mechanism. The free-running tip-speed ratio is about 0.11–0.22 (estimated from V_{oc} through the lab’s generator constant), and the electrical power coefficient at peak, $P_{max}/(\frac{1}{2}\rho A v^3)$, is 0.09–0.27%. The rotor is drag-driven: its torque is the difference between drag on the concave advancing face and the convex returning face. Roughness on the convex face can delay separation and reduce its drag, as on a golf ball. That is a candidate, not a demonstrated mechanism. Lower parasitic torque would also raise V_{oc} , and only a $C_p(\lambda)$ comparison with measured rotor speed can separate the two. The Sandia roughness study circulated in August [1] concerns airfoils at utility-scale Reynolds numbers, where roughness is detrimental; it does not apply to this rotor.

Table 3. Limitations, their effect, and what would resolve them.

Limitation	Effect on the conclusions	Resolved by
Run order equals texture order	Drift would mimic texture; the evidence rules out a large drift, not a few percent (Section 3.4)	Interleaved order with reference re-runs
Rotors not remounted on 1 October	Mounting variation estimated from three earlier runs only (3 degrees of freedom), which widens the intervals	Remount each rotor between runs, on more than one day
Reference printed with 0.2 mm layers	The no-texture comparison mixes texture with layer height	Reprint at 0.1 mm layers
One print per texture level	Mass, wall, edges and the Ra 40 magnet may co-vary with texture	Second print set; weigh blades; record print settings for every set
One small field per surface	Pa and Ra uncertain; texture longer than 1.43 mm not captured	Several fields per set; profiles of 4 mm or more
No rotor speed	Electrical power only; no λ or $C_p(\lambda)$	Lab pulse in parallel, or a Hall sensor
No ambient air or independent wind	Density bounded only by fan current; wind from the fan-speed calibration	Tunnel node and anemometer on every run

6 Next steps

1. **Reprint the no-texture set** (one blade broke) at 0.1 mm layers, as for Ra 20 and Ra 40, and confirm every set’s settings from its slicer (.3mf) file.
2. **Repeat with remounting, in an interleaved order:** remove and remount each rotor between runs, and re-run the reference every few runs.
3. **Extend the surface measurement:** several fields per set, longer profiles, and the convex face near the leading edge.
4. **Add rotor speed and ambient air** to every run, and record one sweep with the lab DAQ in parallel.

6.1 Toward the fault-testing plan

Prof. Jeong asked for a fault-testing plan agreed before the next round. The campaign suggests a starting point.

- **What the rig offers.** An automated sweep takes 8 minutes, with a run-to-run repeatability of 0.6% and a mounting-plus-day variation estimated at 1.5% from three comparisons. The Thévenin decomposition helps separate rotor-side faults, which move V_{oc} , from electrical ones,

which move R_{int} ; rotor-side drag can move R_{int} too. The tunnel node's accelerometer adds vibration spectra.

- **Candidate faults.** A tip mass on one blade (imbalance); a chipped, shortened or missing blade; added bearing drag; an open or high-resistance generator phase.
- **Proposed next step.** A short meeting to agree the fault list, severity levels and acceptance criteria before any fault blades are printed.

References

- [1] B. Wilcox, E. White and D. Maniaci, "Roughness sensitivity comparisons of wind turbine blade sections," Sandia National Laboratories, 2017. OSTI 1404826.

A Per-set-point data

Table 4. Peak electrical power (W), largest 1 s dwell, for every 1 October run. 1 and 2: first and second run on the same mounting. Wind speed is nominal, from the commanded fan speed. The August–September runs are in the data package.

Fan rpm	v m/s	No texture		Ra 20		Ra 40		Ra 80	
		1	2	1	2	1	2	1	2
500	10.2	0.0294	0.0298	0.0313	0.0317	0.0330	0.0348	0.0303	0.0339
600	12.4	0.0658	0.0660	0.0638	0.0633	0.0707	0.0704	0.0676	0.0660
700	14.5	0.1140	0.1175	0.1128	0.1153	0.1197	0.1184	0.1232	0.1218
800	16.6	0.1858	0.1811	0.1808	0.1794	0.1931	0.1969	0.1983	0.2023
900	18.8	0.2753	0.2937	0.2864	0.2644	0.3045	0.3135	0.3181	0.3220
1000	20.9	0.4162	0.4109	0.4158	0.4044	0.4716	0.4671	0.5362	0.5516
1100	23.0	0.6093	0.6031	0.6474	0.6572	0.7309	0.7381	0.8278	0.8240
1200	25.2	0.8894	0.8914	0.9700	0.9642	1.0875	1.0921	1.1261	1.1142
1300	27.3	1.2706	1.2463	1.3247	1.2875	1.4130	1.4123	1.4403	1.4474
1400	29.4	1.5788	1.5975	1.6391	1.6184	1.7898	1.7959	1.8351	1.8511
1500	31.6	1.9799	2.0368	2.0413	2.0236	2.2892	2.2066	2.2762	2.2513
1600	33.7	2.4530	2.4832	2.4978	2.5259	2.7952	2.8414	2.9085	2.8549
1700	35.8	3.0003	3.0273	3.0671	3.0700	3.4574	3.5213	3.5436	3.5516
1800	38.0	3.6640	3.6666	3.7998	3.7517	4.3137	4.2905	4.4798	4.5882

B Methods detail

Peak power. The dwells in which the load tracked its demand with current above zero form the ladder (I_k, P_k) . The raw estimate is $\max_k P_k$. The fit takes the $\arg \max$ and up to two dwells either side, fits $P = a_2 I^2 + a_1 I + a_0$, and reports the vertex if $a_2 < 0$ and it lies inside the window; otherwise the $\arg \max$. This is a part of the rig software’s rule. On the fit basis the changes are +1.6% [−0.3, +3.5], +13.0% [+10.9, +15.2] and +16.2% [+14.1, +18.4] (run-to-run intervals, $\sigma = 0.7\%$), against +1.8%, +12.8% and +16.2% on the $\arg \max$.

Run comparison. Run j of rotor r gives $y_{rj} = \frac{1}{n} \sum_s \ln(P_{js}/P_{\text{ref},s})$ over the $n = 14$ set points common to every run. The pooled standard deviation is $\sigma^2 = \sum_r \sum_j (y_{rj} - \bar{y}_r)^2 / \sum_r (n_r - 1)$. The change between rotors a and b is $\exp(\bar{y}_a - \bar{y}_b) - 1$, with interval $\exp(\bar{y}_a - \bar{y}_b \pm t_{0.975,4} \sigma \sqrt{1/n_a + 1/n_b}) - 1$ for run-to-run repeatability.

Mounting variation. For textured rotor r , x_r is the log-ratio of its 1 October mean to its August–September run, a separate mounting, so $\text{Var}(x) = 2\sigma_m^2 + 1.5\sigma^2$. Treating the three x_r as zero-mean, which counts any common day offset as mounting variation, gives $\sigma_m^2 = (\overline{x^2} - 1.5\sigma^2)/2$ with 3 degrees of freedom. A 1 October comparison of two rotors, each mounted once, then has $\text{SE}^2 = 2\sigma_m^2 + \sigma^2(1/n_a + 1/n_b)$, with $t_{0.975,3}$ (the Satterthwaite approximation gives 2.8 degrees of freedom, which changes no conclusion). The no-texture rotor has no August–September run; its mounting variation is assumed equal to the textured rotors’.

Thévenin fit. At each set point, $V = V_{\text{oc}} - IR_{\text{int}}$ by least squares over all tracking dwells. Each rotor’s curve is the geometric mean of its two runs. The interval on the mean change is a t interval across set points.

Surface parameters. The Keyence export is a height map in millimetres. Each image column is one spanwise profile. Columns with more than 10% missing points are dropped, short interior gaps are filled linearly, a least-squares line is removed, and profiles are cut to the length of the shortest one and re-centred. P_a is the mean absolute deviation of that profile. For R_a , the profile is filtered with the ISO 16610-21 Gaussian ($\alpha = \sqrt{\ln 2/\pi}$), $\lambda_c = 0.25$ mm, and $\lambda_c/2$ is discarded at each end. The layer period is the peak of the mean power spectrum between 60 and 320 μm . Values are means over about 1000 profiles per field. For periodic profiles ISO 4288 selects $\lambda_c = 0.25$ mm for spacings of 0.04–0.13 mm, which fits the 0.1 mm period of $R_a 20$ and $R_a 40$. The 0.2 mm period of the no-texture set would call for 0.8 mm, and the non-periodic $R_a 80$ surface for 2.5 mm, both too long for the 1.43 mm field. R_a here therefore describes texture finer than 0.25 mm for every set, and P_a everything up to the field length.

C Reprocessing of the 27 July export

The file `0727windturbine.csv` holds 127 208 rows at 360 Hz. The channel map below was inferred from the data and should be confirmed against the lab’s wiring.

- **ch1:** DC bus voltage through a 4:1 divider.
- **ch2:** Hall-effect current sensor, 2 A/V about a mid-rail zero.
- **ch3:** once-per-revolution rotor pulse. It tracks cleanly up to 1900 rpm and misses about one revolution in nine at 2000 rpm.

Rebuilding Summary_Table.csv. With $L = \lfloor 127208/17 \rfloor = 7482$, window $k = 0 \dots 15$ covers samples $kL + 1870$ to $kL + 5611$: the middle half of each of the first 16 of 17 equal slices. With $V = 4 \text{ ch1}$ and $I = 2(\text{ch2} - 2.5)$, V_{dc_max} , I_{dc_max} and P_{dc_max} are the separate maxima of V , I and VI over the window. **Measured_RPM** is 60 times the number of rising ch3 edges over the window duration. All 64 tabulated values reproduce to floating-point precision.

What the rebuild shows.

- The current channel carries about 31 mA of white noise, load on or off. At each window’s P_{dc_max} sample the current is 0.095–0.144 A above its 1 s mean. Applied to a stretch with the load disconnected, the recipe returns 1.50 W.
- The load was held in constant current, raised by hand, and off for the first three settings.
- Settings were changed by hand, with dwells of 12 to about 40 s, so fixed windows can straddle a change: the 1300 rpm window contains the step to 1400 rpm.

Reprocessing. Same windows; current zero from ch2 after the load is switched off (2.4963 V); power averaged over a centred 1 s window before taking the maximum.

D Data package and reproducibility

The accompanying archive contains:

- `1_rig_sweeps/`: every rig file used, byte for byte (August–September and 1 October);
- `2_jeong_lab/`: the lab’s tables, raw export and plots, as e-mailed;
- `3_surface_scans/`: the Keyence height maps and screenshots;
- `4_derived/`: the analysis-ready tables behind this report;
- `5_reference/`: the blade geometry;
- `6_figures/` and `7_code/`: the figures and the analysis code.

Table 5. 27 July no-texture test by fan set point: the lab’s `Pdc_max`; the same window reprocessed; mean load current; the rig’s un-textured rotor (1 October); and the ratio of the reprocessed value to it. [†]Window contains the step to 1400 rpm.

Fan rpm	Lab <code>Pdc_max</code> (W)	Reprocessed (W)	Mean current (mA)	Rig, no texture (W)	Ratio
500	0.333	0.008	−1	0.030	—
600	0.533	0.017	0	0.066	—
700	0.508	0.014	0	0.116	—
800	0.607	0.093	15	0.183	—
900	0.768	0.192	28	0.284	0.68
1000	0.918	0.277	40	0.414	0.67
1100	1.016	0.449	69	0.606	0.74
1200	1.340	0.634	101	0.890	0.71
1300 [†]	1.825	1.117	127	1.258	0.89
1400	2.253	1.478	178	1.588	0.93
1500	2.611	1.901	223	2.008	0.95
1600	3.103	2.227	231	2.468	0.90
1700	3.818	2.690	271	3.014	0.89
1800	4.304	3.124	302	3.665	0.85
1900	5.073	3.720	340	—	—
2000	5.602	4.261	348	—	—

`README.md` explains the layout, `DATA_DICTIONARY.md` defines every column, and `MANIFEST.csv` gives each file’s SHA-256 checksum. Running `python3 7_code/build_report.py` regenerates every result, table and data figure in this report from the raw files. The August–September files record fan speed differently from each other; use the commanded speed (`README.md`).